

Homework 05 - Reconstruction & the discrete Fourier transform

Deadline: 2nd test (date to be determined but not earlier than the 9th of April)

The mandatory questions account for 100 points, while the optional questions together contribute up to an additional 50 points.

1. Sketch diagrammatically the steps for sampling and reconstruction of a continuous signal. Explain it!
2. What is the condition for a continuous signal to be reconstructible from its discrete time samples?
3. What is an antialiasing filter? What is its role (benefits & limitations) ?
4. Using the scripts made available to you demonstrate the reconstruction process for a “continuous” signal $x(t)$ with a parabolic spectrum

$$X(f) = \begin{cases} f_0^2 - f^2, & |f| < f_0 \\ 0 & |f| \geq f_0 \end{cases}, \quad f_0 > 0$$

- (a) Create and represent the above function
 - (b) Compute and represent the $x(t)$ signal.
 - (c) Demonstrate the sampling process.
 - (d) Compute the Fourier transform of the sampled signal.
 - (e)  Reconstruct the original signal from the sampled signal and show that the reconstruction is not complete if f_0 is larger than the Nyquist frequency.
5. Show that uniform sampling of a signal leads to a periodic Fourier transform.
 6. What is the period of the Fourier transform of a sampled signal?
 7. Show that a periodic signal has a sampled Fourier transform.
 8. What is the sampling period of the Fourier transform of a periodic signal?
 9.  Derive the discrete Fourier transform $X[k]$ of $x[n]$, a T sampled NT periodic signal. Derive also the expression of the inverse Fourier transform returning $x[n]$ from $X[k]$.
 10. What is the frequency resolution of the discrete Fourier transform of a T sampled signal?
 11. What are the frequency limits (min/max) of the discrete Fourier transform of a T sampled signal?
 12.  Show that

$$\sum_{k=0}^{N-1} e^{\frac{2\pi i}{N}(m-n)k} = \begin{cases} N, & m = n \\ 0, & m \neq n \end{cases}$$

13.  Test that indeed applying subsequently the DFT and its inverse we can restore the original sampled and periodic signal.